

You must show your work to receive credit. Each problem is worth 4 pts unless stated otherwise.

1. Use the function, $f(x) = 4^x$, to complete parts 1a–1d.

a. Graph $f(x)$. Label the y -int and at least one other point.

b. Find the domain of $f(x)$.

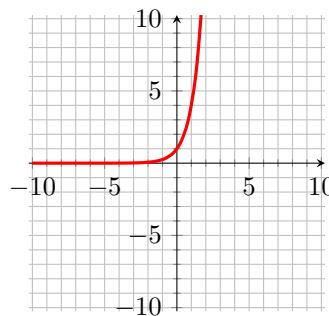
$$(-\infty, \infty)$$

c. Find the range of $f(x)$.

$$(0, \infty)$$

d. Find the horizontal asymptote of $f(x)$.

$$y = 0$$

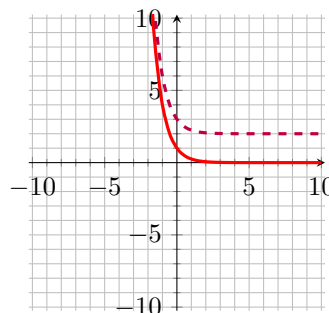


2. Use the function, $f(x) = \left(\frac{1}{4}\right)^x$, to complete parts 2a–2b.

a. Graph $f(x)$. Label the y -int and at least one other point.

b. Find the horizontal asymptote of

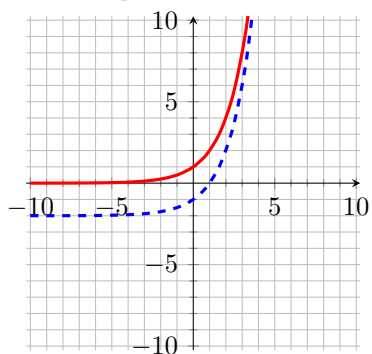
$$g(x) = \left(\frac{1}{4}\right)^x + 2.$$



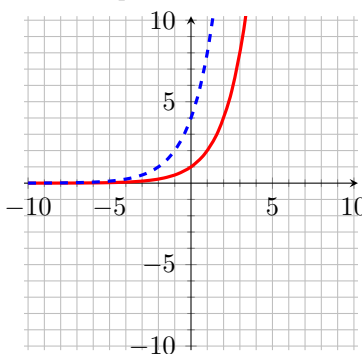
$$y = 2 \text{ (} g(x) \text{ is the dashed curve)}$$

3. Use the graph of b^x in parts 3a–3c to graph the given transformation.

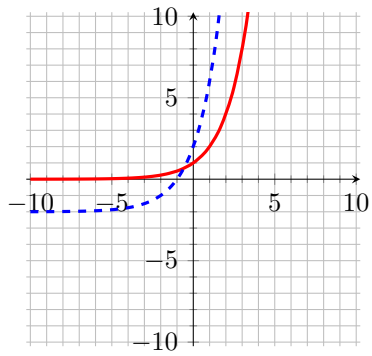
a. Graph $b^x - 2$



b. Graph b^{x+2}



c. Graph $b^{x+2} - 2$



Use the formulas, $A = P(1 + r)^t$, $A = P(1 + \frac{r}{n})^{nt}$, or $A = Pe^{rt}$ to problems 4–6.

4. [6 points] Find the accumulated value of an investment of \$850 at 9% interest compounded quarterly after 20 years.

$$P = 850, r = .09, t = 20, \text{ and } n = 4. \text{ So } A = (1 + \frac{.09}{4})^{4 \cdot 20} \approx \$5040.624$$

5. [6 points] Bob buys a house for \$125,000. The value of the house grows at a rate of 4.5% compounded monthly. Find how much the house is worth after 15 years.

$$P = 125000, r = .045, t = 15, \text{ and } n = 12. \text{ So } A = \left(1 + \frac{.045}{12}\right)^{12 \cdot 15} \approx \$227,087.087$$

6. [6 points] Suppose Jane has \$1000 to invest. Which investment yields the greater return over 10 years: 7% compounded monthly or 6.5% compounded continuously? Justify your answer.

- 7% compounded monthly

$$P = 1000, r = .07, t = 10, \text{ and } n = 12. \text{ So } A = \left(1 + \frac{.07}{12}\right)^{12 \cdot 10} \approx 2009.661$$

- 6.5% compounded continuously

$$P = 1000, r = .065, t = 10. \text{ So } A = 2000e^{.065 \cdot 10} \approx \$1915.541$$

So 7% compounded monthly is better.

7. Write the expression in its equivalent *exponential* form for parts 7a–7c.

a. $\log_2 x = 100$
 $2^{100} = x$

b. $\ln x = 2$
 $e^1 = 2$

c. $\log_x 5 = y$
 $x^y = 5$

8. Write the expression in its equivalent *logarithmic* form for parts 8a–8c.

a. $4^x = 64$

b. $x^{14} = 9$

c. $x^{-3} = \frac{1}{8}$
 $\log_x \frac{1}{8} = -3$

$$\log_4 64 = x$$

$$\log_x 9 = 14$$

9. [3 points each] Evaluate or simplify the expression without using a calculator for parts 9a–9f.

a. $\log_6 6 = 1$

c. $\log_2 2^{12} = 12$

e. $\log_4 16 = 2$ since $4^2 = 16$

b. $e^{\ln 3} = 3$

d. $4^{\log_4 10} = 10$

f. $\log 10$ since $10^1 = 10$

10. Find the domain of the logarithmic function for parts 10a–10b.

a. $\log_{13} x$

Domain: $(0, \infty)$

b. $\ln(x - 3)$

$$x - 3 > 0 \Rightarrow x > 3$$

Domain: $(3, \infty)$